

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: French Pharmaceutical Regulatory Affairs — Document Compliance NLP

Input Ontology definition: Whether the benchmark’s test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

DrBenchmark's 20-task taxonomy spans POS, NER, MCQA, MCC, MLC, intent classification, and STS [Q1, Q18, Q81], which superficially includes the two task types (NER and STS) the deployment requires. However, none of the 20 tasks is designed around EU regulatory submission genres — SmPCs, PILs, CTD modules, or ANSM submission formats. Web search confirmed no DrBenchmark task uses SmPC or CTD text as a compliance-checking genre [WEB-2, WEB-3]. Dataset analysis confirms zero datasets use regulatory documents as the primary evaluation target: QUAERO/emea and MANTRAGSC/fr_emea source from EMA leaflets but only for NER, not STS compliance equivalence. The taxonomy distribution also underweights the two highest-priority task types: only QUAERO, MANTRAGSC, DEFT2021-ner, and E3C contribute token-level NER with partial pharmaceutical relevance, and only DEFT2020 and CLISTER contribute STS. Many other tasks (FrenchMedMCQA, MORFITT, DiaMED, CAS/ESSAI negation-speculation) consume benchmark coverage but are categorically misaligned with the deployment's NER/STS compliance-checking needs.

ID	Evidence
Q1	"It encompasses 20 diversified tasks, including named-entity recognition, part-of-speech tagging, question-answering, semantic textual similarity, and classification." (p.1)
Q18	"A variety of tasks with different requirements and objectives: Part-of-Speech (POS) tagging, Multi-class, Multi-label and Intent classification, Named-Entity Recognition (NER), Multiple-Choice Question-Answering (MCQA), and Semantic Textual Similarity (STS)." (p.2)
Q81	"Table 8 summarizes the results obtained on average by the considered MLMs when aggregating the tasks into one of the five designated categories: POS, NER, MCQA, MCC (Multi-class classification), MLC (Multi-label classification), or STS tasks." (p.8)
WEB-2	DiaMed IAA Table 4 numerical values available in full paper PDF but not surfaced in indexed text (https://aclanthology.org/2024.lrec-main.478/)
WEB-3	DrBenchmark paper makes no mention of French overseas territory vocabulary coverage (https://arxiv.org/abs/2402.13432)

Strengths

- Includes both NER and STS task types relevant to the deployment [Q1, Q18]
- Negation/speculation detection in CAS and ESSAI provides marginal alignment with detecting uncertain or qualified safety claims (CAS-D3, CAS-D4, ESSAI-D6)
- PxCorpus intent classes 'replace' and 'negate' model dosage correction speech acts that are conceptually adjacent to compliance flagging (PXCORPUS-D8, PXCORPUS-D9)
- DrBenchmark is the only large-scale French biomedical NLP evaluation framework available, making it the most informative available proxy for model selection [Q16, Q83]

ID	Evidence
Q1	"It encompasses 20 diversified tasks, including named-entity recognition, part-of-speech tagging, question-answering, semantic textual similarity, and classification." (p.1)
Q16	"To our knowledge, aside the multilingual benchmark BigBIO (Fries et al., 2022) which includes only 4 corpora for French and is initially intended for generative text completion under zero-shot scenario, no large benchmark specialized in the French biomedical field exists." (p.2)

Q18	"A variety of tasks with different requirements and objectives: Part-of-Speech (POS) tagging, Multi-class, Multi-label and Intent classification, Named-Entity Recognition (NER), Multiple-Choice Question-Answering (MCQA), and Semantic Textual Similarity (STS)." (p.2)
Q83	"In this paper, we introduced DrBenchmark, the first large language understanding benchmark tailored for the French biomedical domain." (p.9)
CAS-D3	le diagnostic d'ulcère solitaire du rectum était évoqué à tort et une rééducation recto-sphinctérienne réalisée sans succès. — Combined negation + speculation in clinical narrative — demonstrates label complexity
CAS-D4	il est donc possible que le même résultat clinique aurait été observé chez ce patient sans ajout de la nac. — Speculative treatment outcome claim — relevant to pharmacovigilance reasoning
ESSAI-D6	Des traitements de radiothérapie plus courts sur 3 semaines pour des cancers du sein sans atteinte ganglionnaire se sont montrés tout aussi efficace et ne présentent pas plus d' effets secondaires dans différents essais cliniques sur plusieurs milliers de patientes. — Negation of adverse effects — partially relevant to safety claim checking
PXCORPUS-D8	attention il s'agit de 20 milligrammes et pas 10 milligrammes — Explicit dosage correction speech act
PXCORPUS-D9	remplacer 1 comprimé tous les jours par 1 comprimé en cas d'anxiété — Regimen replacement: daily → conditional

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required deployment categories include: NER over regulatory entity types (INNs, ATC codes, excipients, posology fields, contraindication qualifiers, MedDRA terms) and STS calibrated for regulatory equivalence over SmPC/PIL/CTD text.

Confirmed via deployment context and regulatory framework documentation [WEB-5, WEB-6].

IO-2: Yes — multiple regionally-relevant categories are omitted: no SmPC compliance task, no CTD module task, no ANSM submission format task, no regulatory variation type (IA/IB/II) classification task, and no MedDRA-aligned NER task. Confirmed by web search [WEB-2, WEB-3] and dataset analysis (cross-cutting CRITICAL [IO] finding).

IO-3: Yes — several included tasks are categorically misaligned with the deployment: FrenchMedMCQA (pharmacy exam MCQA, FRENCHMEDMCQA-D11), MORFITT (specialty routing of research abstracts, MORFITT-D1), DiaMED (ICD-10 chapter classification of Sub-Saharan African clinical cases, DIAMED-D9, DIAMED-D11), and CAS/ESSAI negation-speculation tasks (CAS-D12, ESSAI-D8) consume benchmark weight without contributing to NER/STS compliance evaluation.

IO-4: Major content validity gap: the benchmark's task taxonomy is constructed for general French biomedical NLP evaluation, not regulatory compliance. The closest regulatory-adjacent tasks (QUAERO-emea NER, DEFT2020 STS) are framed for clinical/biomedical scoring, not regulatory equivalence judgment [Q22, Q34, Q41].

ID	Evidence
Q1	"It encompasses 20 diversified tasks, including named-entity recognition, part-of-speech tagging, question-answering, semantic textual similarity, and classification." (p.1)
Q18	"A variety of tasks with different requirements and objectives: Part-of-Speech (POS) tagging, Multi-class, Multi-label and Intent classification, Named-Entity Recognition (NER), Multiple-Choice Question-Answering (MCQA), and Semantic Textual Similarity (STS)." (p.2)
Q22	"The first task aims at identifying the degree of similarity within pairs of sentences, from 0 (the less similar) to 5 (the most similar)." (p.3)

Q34	"The dataset covers two text genres (drug leaflets and biomedical titles), consisting of a total of 103,056 words sourced from EMEA or MEDLINE." (p.3)
Q39	"FrenchMedMCQA (Labrak et al., 2022) is a Multiple-Choice Question-Answering (MCQA) dataset for biomedical domain. It contains 3,105 questions coming from real exams of the French medical specialization diploma in pharmacy, integrating single and multiple answers. The first task consists of automatically identifying the set of correct answers among the 5 proposed for a given question. The second task consists of identifying the number of answers (between 1 and 5) supposedly correct for a given question." (p.4)
Q41	"CLISTER (Hiebel et al., 2022) is a French clinical cases Semantic textual similarity (STS) dataset of 1,000 sentence pairs manually annotated by several annotators, who assigned similarity scores ranging from 0 to 5 to each pair. The scores were then averaged together to obtain a floating-point number representing the overall similarity. The objective of this dataset is to develop models that can automatically predict a similarity score that closely aligns with the reference score based solely on the two sentences provided." (p.4)
Q83	"In this paper, we introduced DrBenchmark, the first large language understanding benchmark tailored for the French biomedical domain." (p.9)
MORFITT-D1	La mise en place par cathétérisme d'une valve aortique (TAVI) destinée à traiter les malades ayant un rétrécissement aortique a été une des plus importantes innovations médicales de notre siècle. — TAVI cardiac innovation abstract — formal biomedical French register
MORFITT-D2	Nous avons récemment publié les conclusions d'un nouveau test préclinique portant sur le dépistage analgésique rapide basé sur l'injection intraplaire (i.pl.) d'une solution saline hypertonique à 10 % (HS) chez des souris femelles croisées (CD-1). — Preclinical analgesic screening — pharmacology label but no NER/STS annotation
QUAERO-D2	TYSABRI 300 mg solution à diluer pour perfusion natalizumab Chaque flacon de 15 ml de concentré contient 300 mg de natalizumab (20 mg / ml) — Brand name, INN, dosage, form, concentration co-occurring in regulatory sentence
DIAMED-D6	Monsieur Y. O. âgé de 19 ans a été hospitalisé le 03 février 2012 pour une paraplégie évoluant depuis un mois. — Clinical case narrative genre — not regulatory document genre
ESSAI-D8	Dans ce contexte, l' étude ESMART a pour objectif d' explorer des nouveaux médicaments, seul ou en association, de chercher quelle est la meilleure dose chez l' enfant, adolescent et jeune adulte, et si des anomalies retrouvées ou non dans la tumeur mènent à un avantage thérapeutique chez ces patients. — Investigational framing absent from approved regulatory document language
DIAMED-D9	Communication interventriculaire ischémique... dans le service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Explicitly Burkinabé hospital setting — not metropolitan France
FRENCHMED MCQA-D11	Au cours de la leucémie lymphoïde chronique, le myélogramme montre: Une population de lymphocytes>30%... — MCQA format; no NER labels or STS scores present
DIAMED-D11	Ces douleurs étaient consécutives à une alimentation importante suite à la rupture du jeûn durant le mois de ramadan. — Ramadan context confirms African/North African origin; cultural/geographic mismatch
CAS-D12	l'examen clinique montre un état général conservé. — Neutral label for factual clinical statement — irrelevant to regulatory entity NER
DEFT2020-D22	Ketoderm 2%, gel en récipient unidose peut donc être utilisé au cours de la grossesse." / "Qu'est-ce que ketoderm 2%, gel en récipient unidose et dans quel cas est-il utilisé. — Patient leaflet format, not SmPC Section 4.6 format
WEB-2	DiaMed IAA Table 4 numerical values available in full paper PDF but not surfaced in indexed text (https://aclanthology.org/2024.lrec-main.478/)
WEB-3	DrBenchmark paper makes no mention of French overseas territory vocabulary coverage (https://arxiv.org/abs/2402.13432)
WEB-5	ANSM-specific packaging requirements (CIP codes, Exploitant, List I/II) not reflected in benchmark content (https://www.pharmalex.com/thought-leadership/blogs/the-complex-landscape-of-french-regulatory-affairs/)
WEB-6	Variation types IA/IB/II directly map to threshold-based escalation requirements but are not evaluated in benchmark (https://alhena-consult.com/how-to-market-a-drug-in-france/)

Information Gaps

- Whether MORFITT pharmacology subset or DEFT2021 SUBSTANCE/DOSAGE NER tasks could be repurposed as a partial regulatory NER proxy is not addressed by benchmark documentation

Requires Expert Verification

- Whether French regulatory affairs experts would judge any subset of the 20 tasks as adequate for informing a compliance-checking deployment

Remediation

Gap 1: No task is designed around SmPC, PIL, CTD module, or ANSM submission format compliance checking; the closest proxies (QUAERO/emea NER, DEFT2020 STS) are framed for clinical/biomedical scoring.

Recommendation: Build a complementary internal benchmark of SmPC sections (4.3, 4.4, 4.6, 4.8) and PIL templates with paired version-comparison tasks reflecting Type IA/IB/II variation scenarios. Use this to evaluate models on the actual deployment task structure rather than relying on clinical-case proxies.